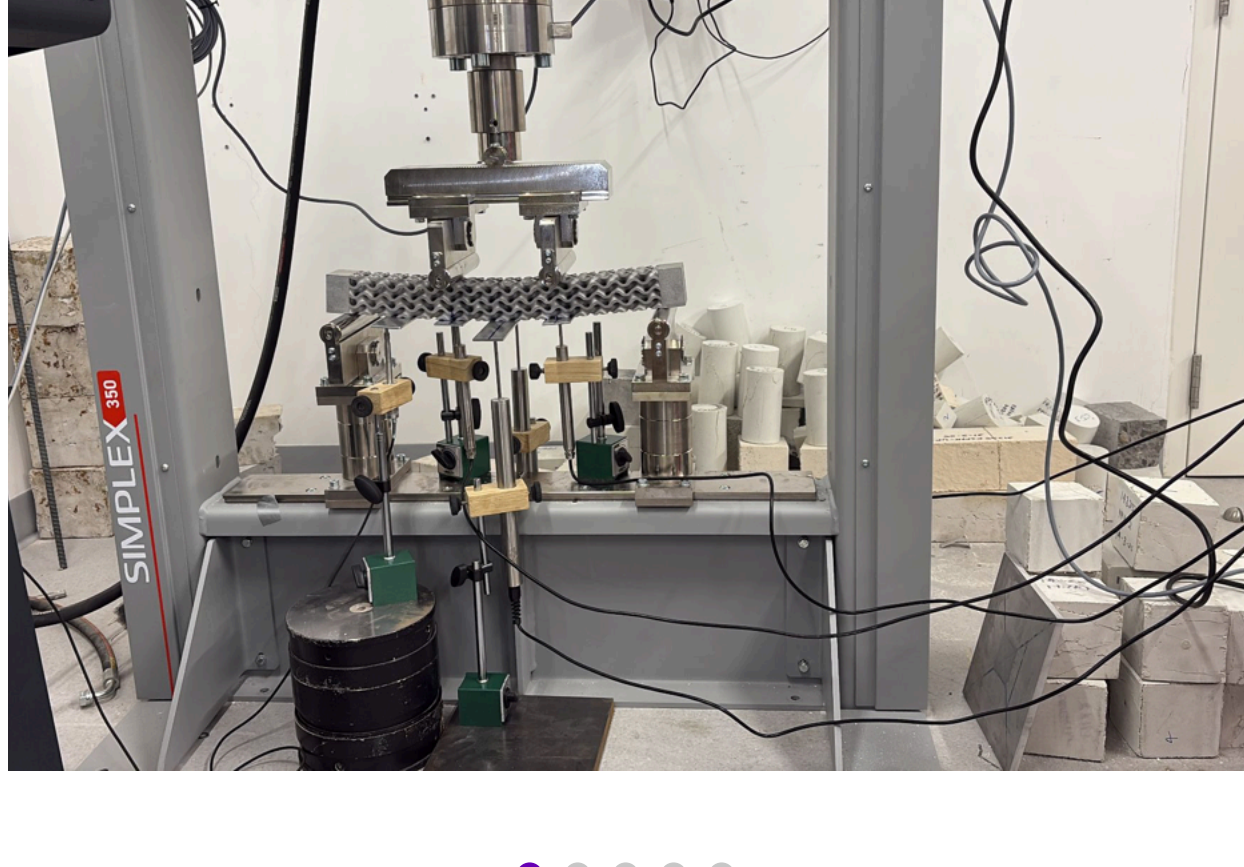


Complexity is no longer manufactured – it is printed. Layer by layer, engineering redefines possibility.

[Current Projects](#) [Past Projects](#) [Home](#)

CURRENT PROJECTS



Large-Scale 3D Printing of Carbon Fiber–Reinforced Composite Metamaterials

Institution: New York University, Abu Dhabi
Duration: Sep 2024 – Present

Overview:

This project focuses on the design, fabrication, and characterization of **large-scale composite metamaterials** using **carbon fiber–reinforced PLA and PPA polymers**. Metamaterials are engineered structures with tailored mechanical and dynamic properties, which are not commonly found in natural materials. By leveraging advanced **additive manufacturing techniques**, this research explores how **geometry, material composition, and scaling** influence the performance of structural components.

Objectives:

- To fabricate **large-scale metamaterial structures** (gyroid and primitive lattices) directly via 3D printing
- To study the **mechanical performance** of structural elements such as cubes and beams under static and dynamic loading
- To understand the **microstructural characteristics** of printed composites and their effect on mechanical behavior
- To develop **numerical models** for predicting stress, deformation, and vibration responses of lattice structures
- To establish **design–processing–property relationships** for engineering lightweight, high-strength, and vibration-sensitive structures

Methodology:

- Design & Fabrication:** Lattice structures including gyroid and primitive topologies were designed and directly 3D-printed using carbon fiber–reinforced PLA and PPA composites. Large-scale cubes and beams were produced to evaluate scalability and structural performance.
- Mechanical Testing:** Quasi-static **compression and flexural tests** were conducted to assess **strength, stiffness, energy absorption**, and failure modes. The results provide insight into how lattice geometry and material reinforcement influence mechanical properties.
- Microstructural Characterization:** Optical microscopy and scanning electron microscopy (SEM) were employed to examine **fiber distribution, interlayer bonding, and printing defects**, ensuring the quality and consistency of fabricated structures.
- Numerical Analysis:** Finite element simulations were performed to predict **stress distribution, deformation patterns, and failure mechanisms**. Results were compared with experimental data to validate models and optimize structural design.
- Dynamic & Vibration Analysis:** Static and dynamic analyses, including modal and frequency response studies, were conducted to evaluate **structural stability, natural frequencies, and vibration behavior**, critical for engineering applications requiring precision and durability.

Significance:

This project demonstrates the potential of **additive manufacturing for high-performance composite metamaterials**. The research provides a framework for designing **lightweight, high-strength structures** with tailored mechanical and dynamic properties, with applications in **aerospace, automotive, robotics, and civil engineering**. The insights gained also contribute to the development of **predictive models for metamaterial performance**, bridging the gap between material science, mechanical engineering, and computational design.

Future Work:

Future directions include **optimization of lattice geometries, exploration of new composite materials, and integration with machine learning models** to predict performance for large-scale engineering applications.

COMPLETED PROJECTS

Additive Manufacturing and Constitutive Modeling of NiTi Shape Memory Alloys

Institution: Khalifa University, Abu Dhabi
Duration: Feb 2020 – Jun 2024

Overview

This project focuses on the **additive manufacturing, characterization, and modeling of NiTi (Nickel–Titanium) shape memory alloys**, with an emphasis on **Triply Periodic Minimal Surface (TPMS)** structures. NiTi alloys exhibit unique **shape memory and superelastic properties**, making them valuable for advanced engineering applications. The work combines **experimental fabrication, microstructural and mechanical characterization, and constitutive modeling** to understand material behavior under various thermal and mechanical conditions.

Objectives

- Fabricate **NiTi alloys and TPMS structures** using **Laser Powder Bed Fusion (LPBF)**
- Perform **microstructural, thermal, and mechanical characterization** of fabricated samples
- Develop **constitutive models** to predict material response, including **fatigue and failure criteria**
- Generate **data-driven insights** for material optimization and publication

Methodology

- Additive Manufacturing:** 3D-printed NiTi samples and TPMS structures using **Laser Powder Bed Fusion (LPBF)**, optimizing printing parameters for structural integrity
- Characterization:** Conducted extensive analysis using **SEM, TEM, XRD, AFM, DSC**, and fatigue testing to examine **microstructure, phase composition, surface morphology, thermal behavior, and mechanical properties**
- Modeling & Simulation:** Performed **numerical simulations and constitutive modeling** to analyze experimental data, predict material behavior, and establish **fatigue failure criteria**
- Research Outputs:** Published **novel findings** in peer-reviewed journals, contributing to the field of **additive manufacturing and smart materials**

Key Skills & Tools

- Additive Manufacturing (LPBF)
- Materials Characterization (SEM, TEM, XRD, AFM, DSC)
- Mechanical Testing and Fatigue Analysis
- Constitutive Modeling and Simulation
- Data Analysis and Scientific Computing

Impact

- Advanced understanding of **NiTi alloy behavior in 3D-printed structures**
- Developed **predictive models** for fatigue and failure in shape memory alloys
- Contributed to the design of **high-performance, smart materials** for engineering and biomedical applications
- Enabled publication of **high-impact research articles** demonstrating innovation in additive manufacturing of smart alloys



Design and Development of Thermo-Responsive and Magnetic Shape Memory Materials

Institution: CSIR Advanced Materials and Processes Research Institute (AMPRI), Bhopal
Duration: Jun 2013 – Aug 2019

Overview

This project focused on the **development of copper-based shape memory materials with enhanced mechanical properties, high transition temperatures, and cost-effective production** in the form of **wires and strips**. Conducted as part of the 12th Five Year Plan of the Government of India, it was a **collaborative effort between CSIR, New Delhi, and CSIR-AMPRI**. The work aimed to design **thermo-responsive and magnetic shape memory materials and devices** for advanced engineering applications.

Objectives

- Develop **copper-based alloys** with superior shape memory and mechanical properties
- Optimize **alloy compositions, heat treatment cycles, and secondary deformation methods**
- Fabricate **wires and strips** suitable for engineering applications
- Perform **microstructural, thermal, and mechanical characterization** to understand material behavior
- Publish findings and establish collaborations for further research

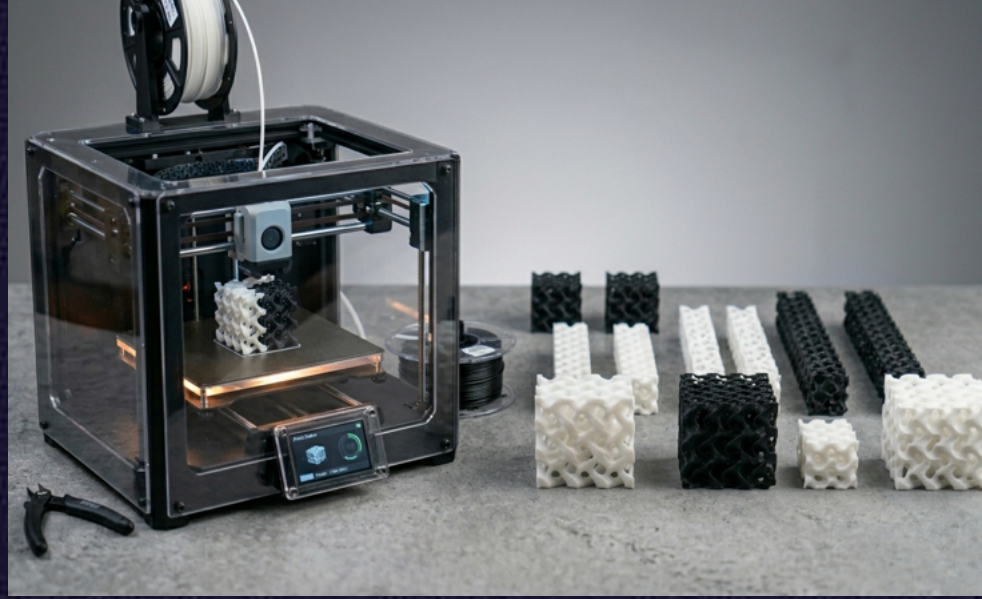
Methodology

- Literature Review & Gap Analysis:** Conducted extensive review to identify knowledge gaps and devise experimental strategies
- Alloy Fabrication:** Synthesized alloys using **vacuum induction melting furnace** and applied secondary deformation techniques such as **hot rolling and tensile processing**
- Characterization Techniques:**
 - Microstructural:** Optical Microscopy, SEM, FESEM
 - Mechanical:** Vickers Hardness, tensile testing
 - Chemical & Phase Analysis:** Optical Emission Spectroscopy, XRD
 - Thermal:** Differential Scanning Calorimetry (DSC)
- Data Analysis:** Experimental results were systematically analyzed to correlate alloy composition, processing, and performance

Key Outcomes & Impact

- Published **five research articles in SCI journals** and delivered **oral/poster presentations** nationally
- Established collaborations with **IIT Madras** for materials research
- Developed **thermo-responsive and magnetic shape memory materials** suitable for engineering devices
- Contributed to **cost-effective production** methods for Cu-based shape memory wires and strips
- Positive reception of research presentations and recognition of project significance in national forums

[Related papers here](#)



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Researcher in Metamaterials, 3D Printing & Shape Memory Materials

Exploring advanced materials for innovative applications

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[Home](#)
[About](#)
[Capabilities](#)
[Projects](#)
[Research & Publications](#)
[Data Science](#)
[Blog](#)
[Contact](#)

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